

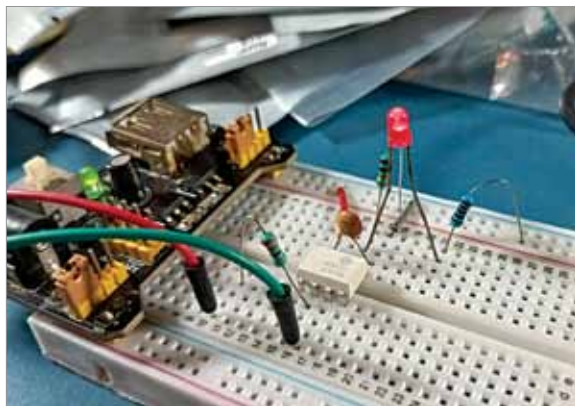


Fig. 4: Author's prototype on breadboard

EFY note. The input signal should have an impedance of a minimum of 50Ω with a rise and fall time of 5ns or greater for the optocoupler to respond properly.

The achievable data rate for any optocoupler depends on how the output is loaded and how it is affected by temperature. Additionally, optocouplers vary widely in many respects, including how devices are characterised from one manufacturer to another.

Using a 20kHz PWM (5V) signal, initial testing with the 6N137 (onsemi device) on a breadboard setup are shown in Fig. 4. For a quick visual indication, a 3mm red LED with series resistor R4 is also wired across the output pull-up resistor ($4.7k\Omega$, R2)—refer to the dotted box in Fig. 3.

The output pulses from a PWM signal generator module are used to drive input of the 6N137 optocoupler. However, most microcontroller I/O pins are capable of delivering at least a few milliamperes of current, so driving the 6N137 optocoupler directly via a current-limiting resistor is not expected to cause issues.

The working of the module is straightforward. First, connect the 5V DC supply across CON3. Then, apply the PWM signal to the input connector CON1. The reciprocal output will be available across CON2, based on the truth table.

For manual testing, the logic behaviour can be verified by applying 5V to the input and enable pins, then observing the output across CON2 using a multimeter. **EFY**



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